Telecom Customer Churn Analytics

Uncovering Retention Drivers Across Indian Telecom Markets

Sector

Telecommunications

Course

DVA Capstone 2 | April 2026

Institute

Newton School of Technology, Rishihood University

Team — Section C, Group 5

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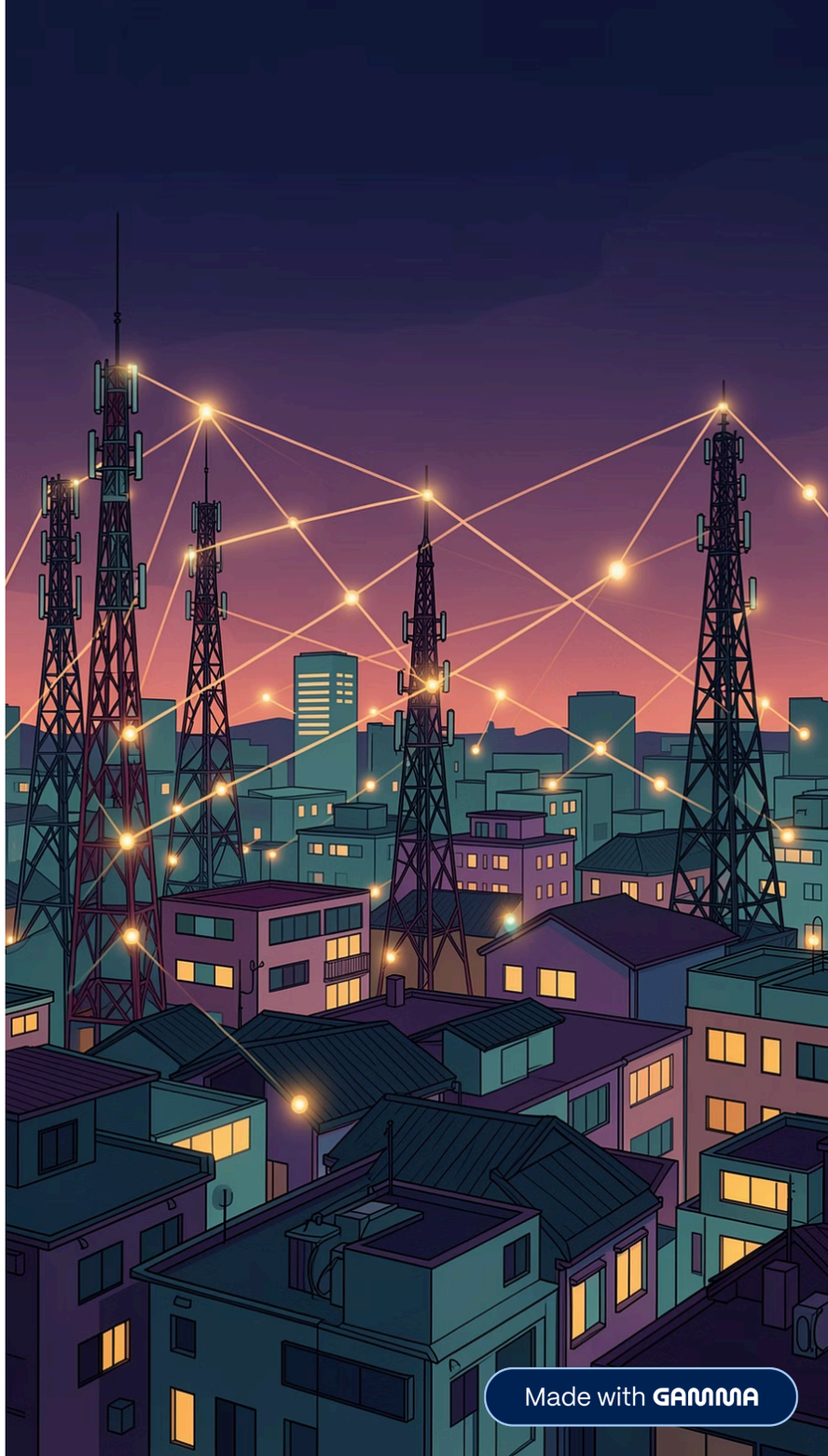

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Context & Problem Statement

Why This Matters

India's telecom sector serves over **1.1 billion subscribers**, making it one of the world's largest. With Airtel, Jio, BSNL, and Vodafone locked in fierce competition, customer retention is mission-critical.

⚠ Acquiring a new customer costs **5–7× more** than retaining an existing one.

Core Business Question

> "Which customer segments are most at risk of churning, and what operational interventions can reduce churn by at least 10% within 6 months?"

Objective

Build an end-to-end analytics pipeline identifying churn patterns across **4 operators** and **28 states** — delivering actionable, data-backed retention strategies.

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Data Engineering

ETL PIPELINE

Dataset at a Glance

243K

Customer Records

4

Operators Covered

28

States | 96 Cities

4

Years of Data

Jan 2020 – Dec 2023

Source: Synthetic Indian Telecom Dataset (Kaggle) | 19 raw columns

Key Cleaning Steps

01

Standardised

Column names to snake_case; dates to ISO format (YYYY-MM-DD)

02

Corrected

Negative values in calls, SMS, data via absolute transformation

03

Normalised

Salary units (lakhs vs. absolute → all converted to ₹); state name abbreviations expanded

04

Dropped

Irrelevant columns (pincode) and duplicate/null Customer IDs

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KPI Framework

Eight KPIs anchor the analytics pipeline — from primary retention health metrics to granular segmentation signals.

Churn Rate

% of customers who left in a period — the primary health metric for retention.

Retention Rate

% who stayed — measures loyalty and plan stickiness over time.

Avg Data Usage

Mean monthly GB per customer — proxy for engagement and plan value.

Usage Score

Weighted composite (0.4×calls + 0.1×SMS + 0.5×data) — holistic engagement index for segmentation.

Avg Annual Income

Mean estimated salary — segments premium vs. price-sensitive users.

Churn by Segment

Churn rate sliced by age group, operator, and state — pinpoints high-risk sub-populations.

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Key EDA Insights

Uniform Cross-Operator Churn (19.86%–20.37%)

All four operators show nearly identical rates — pointing to a **systemic, market-wide problem**, not operator-specific failure.

Elderly Customers: Highest Absolute Churn

12,999 churned customers aged 65+ — the single largest segment — suggesting unmet needs around simplicity, support, and pricing.

Engagement Score Has Zero Predictive Power

Churned (2,529) vs. retained (2,526) — essentially identical. **Current engagement KPIs are unfit for churn prediction.**

State-Level Hotspots Exist

Jharkhand (21.12%), Karnataka (20.71%), and Mizoram (20.65%) lead nationally — signalling localised service or pricing issues.

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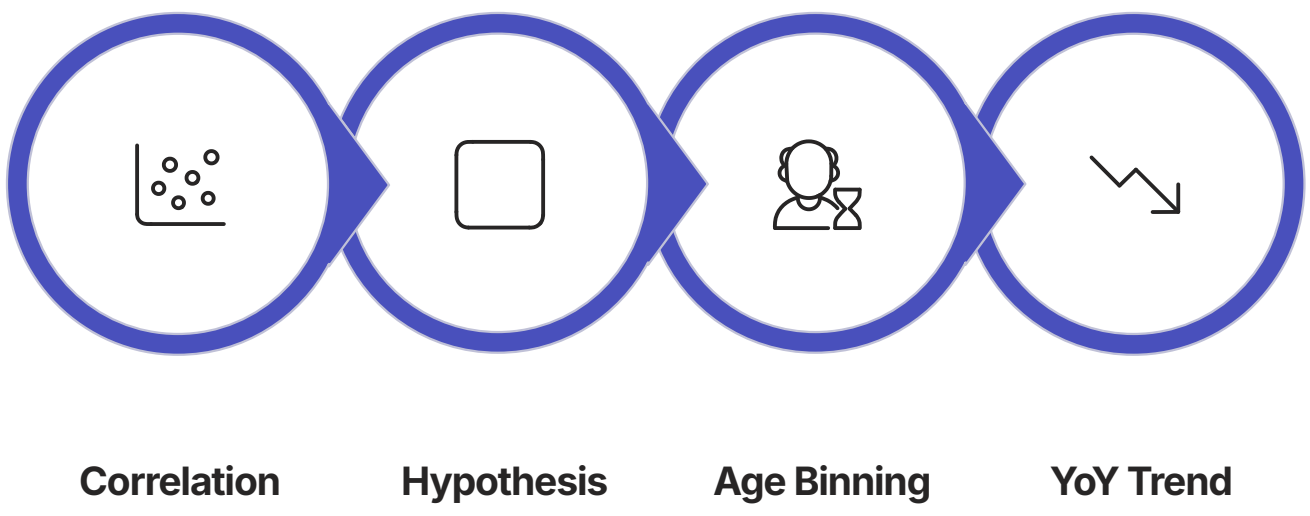
Advanced Analysis

STATISTICAL FINDINGS



Key Findings

Young Adults

+5% churn probability above base rate; Elderly highest in absolute volume due to population size.

Operator Risk

BSNL shows highest weighted churn probability; Jio customers are lowest risk (−8% relative).

2023 Trend

Only 4,917 events recorded — partial-year data. Full-year projection: **15,000–16,000 events**, consistent with prior years.

Near-zero correlation of income & engagement with churn confirms: churn is driven by **external factors** — competitive offers, network perception, plan dissatisfaction.

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Tableau Dashboard Overview

PLATFORM: TABLEAU PUBLIC

1

Executive Market Overview

Total customers, churn KPIs, operator split, India state map

2

Churn Analysis

Year-wise trend, gender breakdown, age group churn rates

3

Engagement Distribution

Usage score distribution, calls vs. data scatter plots

4

Customer Segmentation

Age group profiles, income bands, state-level segmentation

Interactive Filters Available

- Telecom Operator (Airtel / BSNL / Jio / Vodafone)
- State and Region
- Age Group
- Registration Year
- Churned vs. Retained Customer Toggle

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Top Insights

🌐 Market Problem, Not Operator Problem

All operators within a 0.5% churn band. Structural forces — pricing wars and number portability — are dominant.

👴 Elderly = Highest Volume Risk

12,999 churned customers aged 65+. This group is underserved by digital-first strategies and needs dedicated programs.

📍 Geography > Demographics

Jharkhand, Karnataka, Mizoram outpace the national rate. State-specific interventions yield faster ROI than blanket campaigns.

📊 Engagement KPIs Are Broken

Composite usage score shows zero churn correlation. Retention teams need an RFM model — not this metric.

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Recommendations & Impact

Elderly Retention Program
High priority, Med feasibility
Recover 2k-3k customers

State-Level Service Fix
High priority, High feasibility
Reduce regional churn 1-2%

RFM Model for Engagement
Very High priority, Med Met feasibility
Predict churn 30-60d early

Competitive Pricing Review
High priority, High feasibility
Reduce righf churn

24-Month Loyallty Trigger
Med priority, High feasibility
Reduce cliff churn 5-8%

Overall Target

Reduce national churn from **20.05% → below 18%** within 6 months through combined implementation.

Priority Matrix

Recommendation	Priority	Feasibility
RFM Churn Prediction Model	Very High	Medium
Elderly Retention Program	High	Medium
State-Level Intervention	High	High
Competitive Pricing Review	High	High
24-Month Loyalty Trigger	Medium	High

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Limitations, Next Steps & Closing

Limitations to Acknowledge

- **Synthetic Dataset**

  Kaggle simulation — real-world distributions may differ significantly.

- **No Causal Data**

  We identify *who* churns, not *why* — no NPS scores or exit surveys available.

- **Incomplete 2023 Data**

  Only 4,917 events vs. ~14,000+ in prior years — partial year capture.

- **No Plan / Pricing Data**

  Dataset lacks monthly bill, plan type, and promotional history — likely the strongest churn predictors.

Phase 2 Roadmap

01

Predictive ML Model

Logistic regression / gradient boosting classifier using RFM + plan features

02

Real Operator Data Partnership

Collaborate with telecoms for actual CRM and billing datasets

03

Live Churn Alert Dashboard

Tableau upgraded to real-time connection with automated churn risk alerts

04

NLP on Complaints + Cohort Analysis

Social media & support tickets to uncover *why* customers leave

This project delivered a full analytics pipeline across **243,553 records** — proving churn in Indian telecom is a **systemic market phenomenon**, not an operator failure.

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